STATISTICS 652

LECTURE 12 - 2026

- REVIEW AND COMPLETION OF LECTURE 11
- COLLECT HW SET 2

UNDER CONDITIONS OF LEMMA 11.10.ii

WALD-TYPE TEST FOR $H_0: \underset{\sim}{\theta} = \underset{\sim}{\theta}_0$ at approximate level $\alpha$

$$n(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0)' \hat{\mathcal{I}}_n (\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0) > \chi^2_{k,\alpha}$$ REJECTION REGION

WALD-TYPE APPROXIMATE $(1-\alpha)\times 100\%$ CONFIDENCE ELLIPSE FOR $\underset{\sim}{\theta}$

$$\left\{ \underset{\sim}{\theta} \in \mathbb{R}^q \;\middle|\; (\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta})' \hat{\mathcal{I}}_n (\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}) \le \chi^2_{k,\alpha}/n \right\}$$

UNDER CONDITIONS OF LEMMA 11.10.i

SCORES-TYPE TEST FOR $H_0: \underset{\sim}{\theta} = \underset{\sim}{\theta}_0$ at approximate level $\alpha$

$$n(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0)' \mathcal{I}^{-}_{\underset{\sim}{\theta}_0} (\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0) > \chi^2_{k,\alpha}$$ REJECTION REGION

NOTES: - SCORES-TYPE CONFIDENCE REGION WOULD HAVE A COMPLICATE FORM AND WOULD NOT BE AN ELLIPSE.
- WALD'S AND SCORE'S TEST USUALLY REFER TO $\hat{\underset{\sim}{\theta}}_n$ BEING MLE.
- CAN BE GENERALIZED TO $H_0: \underset{\sim}{\theta} \in \Theta_0$ BASED ON $\hat{\underset{\sim}{\theta}}_n - \hat{\underset{\sim}{\theta}}_{0,n}$.

APPLICATION: $\underset{\sim}{X} \sim \text{MULTINOMIAL}_q(n, \underset{\sim}{\pi})$, $\underset{\sim}{\pi} \in \mathbb{R}^q$, $\underset{\sim}{1}'\underset{\sim}{\pi} = 1$

RECALL: $(q-1)$ PARAMETER FULL RANK EXPONENTION FAMILY

MLE: $\hat{\underset{\sim}{\pi}} = \frac{1}{n}\underset{\sim}{X}$ RECALL: $\underset{\sim}{X} \in \mathbb{R}^q$, $\underset{\sim}{1}'\underset{\sim}{X} = n$

EXPRESS $\hat{\underset{\sim}{\pi}} = \frac{1}{n}\underset{\sim}{X} \sim \bar{\underset{\sim}{Y}}$ where $\underset{\sim}{Y}_1, \ldots, \underset{\sim}{Y}_n$ iid $\underset{\sim}{Y} \sim \text{MULTINOMIAL}_q(1, \underset{\sim}{\pi})$.

NOW, $E[\underset{\sim}{Y}] = \underset{\sim}{\pi}$ AND $\text{VAR}(\underset{\sim}{Y}) = \Sigma_{\underset{\sim}{\pi}} = \text{diag}(\underset{\sim}{\pi}) - \underset{\sim}{\pi}\underset{\sim}{\pi}'$

SINCE $Y_j \sim \text{BERNOULLI}(\pi_j)$ and so $E[Y_j] = \pi_j$

$\text{VAR}(Y_j) = \pi_j - \pi_j^2$ and $\text{cov}(Y_j, Y_m) = E[Y_j Y_m] - \pi_j \pi_m = -\pi_j \pi_m$ $j \neq m$

SO, BY WLLN $\hat{\underset{\sim}{\pi}} \to_p \underset{\sim}{\pi}$ and by CLT,

$\sqrt{n}(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi}) \to_d N_q(\underset{\sim}{0}, \Sigma_{\underset{\sim}{\pi}})$ with $\text{rank}(\Sigma_{\underset{\sim}{\pi}}) = q-1$ (HW: VERIFY)

THUS, $T_n = n(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi})\, \Sigma_{\underset{\sim}{\pi}}^{-}\, (\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi}) \to_d \chi^2_{q-1}$.

NOTE: $\Sigma_{\underset{\sim}{\pi}} \underset{\sim}{1} = \underset{\sim}{0}$, and for $\underset{\sim}{a} \propto \underset{\sim}{1}$, $\underset{\sim}{a}' \Sigma_{\underset{\sim}{\pi}} \underset{\sim}{a} > 0$

SINCE $\underset{\sim}{a}' \Sigma_{\underset{\sim}{\pi}} \underset{\sim}{a} = \text{VAR}(\underset{\sim}{a}'\underset{\sim}{Y})$ with $P(\underset{\sim}{a}'\underset{\sim}{Y} = a_j) = \pi_j$ $j = 1, \ldots, q$

HENCE $\text{RANK}(\Sigma_{\underset{\sim}{\pi}}) = q-1$,

NOW: $\Sigma_{\underset{\sim}{\pi}} = D^{1/2}\left(I - D^{-\frac{1}{2}} \underset{\sim}{\pi}\underset{\sim}{\pi}' D^{-\frac{1}{2}}\right) D^{1/2}$ where $D = \text{diag}(\underset{\sim}{\pi})$.

$= D^{\frac{1}{2}}\left(I - \underset{\sim}{b}\underset{\sim}{b}'\right) D^{1/2}$, where $\underset{\sim}{b} = D^{-\frac{1}{2}} \underset{\sim}{\pi} = \begin{pmatrix} \sqrt{\pi_1} \\ \vdots \\ \sqrt{\pi_q} \end{pmatrix}$

with $\underset{\sim}{b}'\underset{\sim}{b} = 1$. HENCE $(I - \underset{\sim}{b}\underset{\sim}{b}')$ IS IDEMPOTENT AND SYMMETRIC with RANK = TRACE $= q-1$

NOW: $\Sigma_{\underset{\sim}{\pi}}^{-} = ?$ COMPLICATED FORM, BUT $(I - \underset{\sim}{b}\underset{\sim}{b}')^{-} = (I - \underset{\sim}{b}\underset{\sim}{b}')$
HW: SHOW

HOWEVER, $\underset{\sim}{D}^{-\frac{1}{2}}(I-\underset{\sim}{b}\underset{\sim}{b}')\underset{\sim}{D}^{-\frac{1}{2}} = \underset{\sim}{D}^{-\frac{1}{2}}(I-\underset{\sim}{b}\underset{\sim}{b}')\underset{\sim}{D}^{-\frac{1}{2}}$

$= \underset{\sim}{D}^{-1} - \underset{\sim}{D}^{-1}\underset{\sim}{\pi}\underset{\sim}{\pi}'\underset{\sim}{D}^{-1} = \underset{\sim}{D}^{-1} - \underset{\sim}{1}\underset{\sim}{1}'$

SCORE'S $\chi^2$

SO $T_n = n(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi})'(\underset{\sim}{D}^{-\frac{1}{2}}(I-\underset{\sim}{b}\underset{\sim}{b}')\underset{\sim}{D}^{-\frac{1}{2}})(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi}) \to_d \chi^2_{q-1}$

SIMPLIFYING: $T_n = n(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi})'\underset{\sim}{D}^{-1}(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi})$ SINCE $\underset{\sim}{1}'(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi}) = 0$

$= \sum_{j=1}^{q} n(\hat{\pi}_j - \pi_j)^2/\pi_j = \sum_{j=1}^{q}(O_j - E_j)^2/E_j$

where $O_j = n\hat{\pi}_j = X_j$ and $E_j = n\pi_j = E(X_j)$. NOTE: $T_n$ IS PEARSON'S $\chi^2$

NOTE: WALD'S $\chi^2$ USES $\hat{\Sigma} = \text{diag}(\hat{\underset{\sim}{\pi}}) - \hat{\underset{\sim}{\pi}}\hat{\underset{\sim}{\pi}}'$ WHICH GIVES

$T_n^* = n(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi})'\hat{\underset{\sim}{D}}^{-\frac{1}{2}}(I - \hat{\underset{\sim}{b}}\hat{\underset{\sim}{b}}')\hat{\underset{\sim}{D}}^{-\frac{1}{2}}(\hat{\underset{\sim}{\pi}} - \underset{\sim}{\pi}) = \sum_{j=1}^{q}(O_j - E_j)^2/O_j \to_d \chi^2_{q-1}$.

## INFORMATION MATRIX.

REGULAR PARAMETRIC MODEL

$\mathcal{F} = \{f_{\underset{\sim}{\theta}} : \underset{\sim}{\theta} \in \Theta\}$ $\Theta \subset \mathbb{R}^q$, $\underset{\sim}{X} \sim f_{\underset{\sim}{\theta}}$, $\underset{\sim}{X} \in \mathbb{R}^n$

UNDER REGULARITY CONDITION

$$I(\underset{\sim}{\theta}; \underset{\sim}{X}) = E_{\underset{\sim}{\theta}}\left[\{\nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta})\}\{\nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta})\}^T\right] > 0$$

where $\nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta}) = \partial \log f(\underset{\sim}{X}; \underset{\sim}{\theta})/\partial \underset{\sim}{\theta}$ IS THE $(q \times 1)$ gradiant with respect to $\underset{\sim}{\theta}$.

OTHER FORMS.

- $I(\underset{\sim}{\theta}; \underset{\sim}{X}) = \text{VAR}\left[\nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta})\right]$ SINCE $E_{\underset{\sim}{\theta}}\left[\nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta})\right] = \underset{\sim}{0}$

- $I(\underset{\sim}{\theta}; \underset{\sim}{X}) = -E_{\theta}\left[\nabla^2 \log f(\underset{\sim}{X}; \underset{\sim}{\theta})\right]$ where

$$\nabla^2 \log f(\underset{\sim}{X}; \underset{\sim}{\theta}) = \left[\frac{\partial^2 \log f(\underset{\sim}{X}; \underset{\sim}{\theta})}{\partial \theta_i \partial \theta_j}\right]_{i,j=1,\ldots,q}$$

THE $q \times q$ IS HESSIAN MATRIX WITH RESPECT TO $\underset{\sim}{\theta}$,

ONE-ONE REPARAMETERIZATION: $\underset{\sim}{\gamma} = g(\underset{\sim}{\theta})$ with $g: \mathbb{R}^q \to \mathbb{R}^q$ being 1-1 and differentiable, then

$$I_o(\underset{\sim}{\gamma}; \underset{\sim}{X}) \equiv I_o(g(\underset{\sim}{\theta}); \underset{\sim}{X}) = \left[\nabla g(\underset{\sim}{\theta})\right]^{-1} I(\underset{\sim}{\theta}, \underset{\sim}{X}) \left\{\left[\nabla g(\underset{\sim}{\theta})\right]^{-1}\right\}^T$$

WHERE $\nabla g(\underset{\sim}{\theta}) = \left[\nabla g_1(\underset{\sim}{\theta}) \cdots \nabla g_q(\underset{\sim}{\theta})\right]$ IS $q \times q$.

FULL RANK

$q$-PARAMETER EXPONENTIAL FAMILY with NATURAL PARMETER $\underset{\sim}{\eta} \in \mathbb{R}^q$, $I(\underset{\sim}{\eta}; \underset{\sim}{X}) = -\nabla^2 d_o(\underset{\sim}{\eta})$

NOTE: - FOR $\gamma = g(\underset{\sim}{\theta})$, $g: \mathbb{R}^q \to \mathbb{R}$, $I(\gamma; \underset{\sim}{X})$ IS NOT WELL DEFINED, SO e.g.

- $I(\underset{\sim}{a}'\underset{\sim}{\theta}; \underset{\sim}{X})$ OR $I(\theta_1; \underset{\sim}{X})$ IS NOT WELL DEFINED

- $I(\theta_1; \underset{\sim}{X})$ MAY BE USED TO DENOTE THE MARGINAL INFORMATION HOLDING $\theta_2, \ldots, \theta_q$ FIXED.

## MULTIVARIATE INFORMATION INEQUALITY

RECALL $\underset{\sim}{X} \in \mathbb{R}^n$, $\underset{\sim}{\theta} \in \mathbb{R}^q$. FOR $\hat{\underset{\sim}{\theta}} = \underset{\sim}{H}(\underset{\sim}{X})$ with $E(\hat{\underset{\sim}{\theta}}) = \underset{\sim}{\theta}$,

$$\text{VAR}(\hat{\underset{\sim}{\theta}}) \geq I(\underset{\sim}{\theta}; \underset{\sim}{X})^{-1}. \quad \text{(LOEWNER ORDERING)}$$

EXAMPLE: $\underset{\sim}{X}_1, \ldots, \underset{\sim}{X}_n$ iid $N_2(\underset{\sim}{\mu}, \Sigma)$ with known $\Sigma > 0$.

VERIFY: $I(\underset{\sim}{\mu}; \underline{\bar{X}}) = n\Sigma^{-1}$, $E(\bar{\underset{\sim}{X}}) = \underset{\sim}{\mu}$ and $\text{VAR}(\bar{X}) = \frac{1}{n}\Sigma$

AND SO $\text{VAR}(\bar{\underset{\sim}{X}}) = I(\underset{\sim}{\mu}; \underline{\bar{X}})^{-1}$.

## PROOF OF INEQUALITY

CONSIDER $h: \mathbb{R}^n \to \mathbb{R}$ and let $\psi(\underset{\sim}{\theta}) = E_{\underset{\sim}{\theta}}[h(\underset{\sim}{X})]$, $\psi: \mathbb{R}^q \to \mathbb{R}$

ANALOGOUS TO DERIVATIONS IN LECTURE 6,

$$\nabla\psi(\underset{\sim}{\theta}) = E\left[h(\underset{\sim}{X}) \nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta})\right] = \text{cov}\left(h(\underset{\sim}{X}), \nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta})\right) \quad q \times 1$$

and $\underset{\sim}{a}^T \nabla\psi(\underset{\sim}{\theta}) = \text{cov}(h(\underset{\sim}{X}), \underset{\sim}{a}^T \nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta}))$

$$\Rightarrow \left[\underset{\sim}{a}^T \nabla\psi(\underset{\sim}{\theta})\right]^2 = \left\{\text{cov}(h(\underset{\sim}{X}), \underset{\sim}{a}' \nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta}))\right\}^2$$

$$\leq \text{VAR}(h(\underset{\sim}{X}))\,\text{VAR}(\underset{\sim}{a}^T \nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta})) \quad \text{by CAUCHY-SCHWARTZ}$$

$$= \text{VAR}(h(\underset{\sim}{X}))\; \underset{\sim}{a}^T \text{VAR}(\nabla \log f(\underset{\sim}{X}; \underset{\sim}{\theta}))\; \underset{\sim}{a}$$

$$= \text{VAR}(h(\underset{\sim}{X}))\; \underset{\sim}{a}^T I(\underset{\sim}{\theta}, \underset{\sim}{X})\; \underset{\sim}{a}$$

CHOOSE $\underset{\sim}{a} = I(\underset{\sim}{\theta}, \underset{\sim}{X})^{-1} \nabla\psi(\underset{\sim}{\theta}) \Rightarrow$

$$\left[\nabla\psi(\theta)^T I(\theta;X)^{-1}\nabla\psi(\theta)\right]^2 \le \mathrm{VAR}\left(h(X)\right)\nabla\psi(\theta)^T I(\theta;X)^{-1}\nabla\psi(\theta)$$

$$\Rightarrow \mathrm{VAR}\left(h(X)\right) \ge \nabla\psi(\theta)^T I(\theta;X)^{-1}\nabla\psi(\theta)$$

CONSIDER FOR $b \in \mathbb{R}^2$, $h(X) = b'\hat{\theta} \Rightarrow \psi(\theta) = b'\theta$ and $\nabla\psi(\theta) = b$.

HENCE $b^T \mathrm{VAR}(\hat{\theta})\, b = \mathrm{VAR}(b'\hat{\theta}) \ge b^T I(\theta;X)^{-1} b$ any $b \in \mathbb{R}^2$

$\Rightarrow \mathrm{VAR}(\hat{\theta}) \ge I(\theta;X)^{-1}$. □

MORE GENERALLY,

COROLLARY 12.1. FOR $X \in \mathbb{R}^n$, $\theta \in \mathbb{R}^q$ and $H: \mathbb{R}^n \to \mathbb{R}^p$

$$\mathrm{VAR}(H(X)) \ge \nabla\psi(\theta)^T I(\theta;X)^{-1}\nabla\psi(\theta)$$

WHERE $\psi(\theta) = E_\theta[H(X)]$ and $\nabla\psi(\theta) = \left[\frac{\partial H_j(X)}{\partial\theta_i}\right]_{q\times p}$.

COROLLARY 12.2. EQUALITY IS OBTAINED IN 12.1 IFF $H(N) = T(\psi)$

IS A NATURAL SUFFICIENT STATISTIC FOR A $k = q$
FULL RANK
PARAMETER EXPONENTIAL FAMILY.

RECALL: $T(\psi)$ IS WELL DEFINED UP TO AFFINE TRANSFORMATIONS